

PROJECT SUMMARY

Accuflow Accounting Solutions — CFO Lens

AI-Assisted Management Accounts Review
& Variance Analysis Platform

APPLICATION TITLE

Accuflow Accounting Solutions CFO Lens

SYSTEM DESIGNATION

AI-Assisted Management Accounts Review & Variance Analysis Platform

TARGET USER PERSONA

CFOs, Financial Controllers, Management Accountants, Business Founders

RELEASE VERSION

1.0.0 Production

"Code Calculates. AI Interprets."

Confidential — Internal Product Documentation

1. Executive Overview

Accuflow Accounting Solutions CFO Lens is a specialized, production-ready financial analytics and executive advisory application. It bridges the gap between raw accounting software exports (QuickBooks Online, Xero, Sage, or custom trial balances) and board-level financial intelligence.

Traditional accounting reporting suffers from two common failure modes:

- **Manual Spreadsheet Fatigue** — controllers spend hours re-keying data into Excel templates, calculating variances, and drafting commentary.
- **AI Arithmetic Hallucination** — generic LLMs frequently fail when asked to calculate financial math, fabricating balances or confusing debits and credits.

Accuflow solves this by enforcing an uncompromising architectural doctrine:

"Code Calculates. AI Interprets." — All mathematical sums, gross margin percentages, month-on-month differences, and anomaly thresholds are computed deterministically in verified TypeScript code. The AI model receives only certified mathematical results and focuses exclusively on high-level operational diagnosis, strategic context, and board-ready recommendations.

2. Core Functional Modules

MODULE 1

Data Ingestion & Quality Audit

UploadView.tsx

- **Dual Format Support:** Ingests multi-period P&L exports in .xlsx, .xls, and .csv formats directly from QuickBooks Online, Xero, or bespoke enterprise ERPs.
- **Automated Period Detection:** Intelligently identifies calendar months (e.g., Jan 2026, Feb 2026, Mar 2026) regardless of column ordering.
- **Data Quality Scoring (0–100%):** Automatically audits uploaded spreadsheets for missing account titles, irregular character encodings, duplicate account rows, and sign inconsistencies.
- **One-Click Demo Mode:** Instant toggle between real client data and a pre-loaded 6-month corporate demo dataset (Apex Dynamics Ltd).

MODULE 2

Chart of Accounts Normalization

MappingView.tsx

- Standardizes diverse vendor account names into six standard GAAP/IFRS categories: Revenue / Turnover; Cost of Goods Sold (COGS) / Direct Costs; Operating Expenses (Overheads / SG&A); Other Income; Other Expense; Taxation.
- **Machine-Assisted Mapping:** Uses keyword heuristic algorithms with visual confidence ratings (High, Medium, Low).
- **Interactive Re-Classification:** Allows finance teams to adjust any classification on the fly, instantly recalculating all downstream P&L metrics and variances.

MODULE 3

Multi-Period Profit & Loss Statement

PLAnalysisView.tsx

- Generates clean, comparative multi-month P&L tables.
- Computes standard financial milestones: Total Turnover / Sales; Cost of Sales & Direct Costs; **Gross Profit & Gross Margin (%)**; Operating Overhead Breakdown; **Operating Profit (EBITDA) & Operating Margin (%)**; Net Profit / Net Margin (%).
- Interactive account drill-down with visual variance flags.

MODULE 4

Deterministic Variance & Materiality Analysis

VarianceView.tsx

- Evaluates Month-on-Month (MoM) performance across both currency (£) and percentage (%) dimensions.
- **Configurable Materiality Filters:** Users can customize both absolute value thresholds (default: £1,000) and percentage change thresholds (default: 10%).
- **Favorable vs. Adverse Classification:** Accurately reverses accounting signs for expense accounts (e.g., an increase in an expense is flagged as Adverse/Unfavorable, while an increase in revenue is Favorable).

MODULE 5

Rule-Based Red Flag & Anomaly Detection

RedFlagsView.tsx

- **Gross Margin Compression:** Flags drops exceeding 200 basis points (2.0%).
- **Expense Run-Rate Expansion:** Detects periods where overhead growth outpaces revenue growth.
- **Sudden Cost Surges:** Identifies line items spiking by >50% and >£2,000.
- **New Account Outliers:** Alerts controllers to previously zero-balance accounts that suddenly book substantial charges.

MODULE 6

Multi-Month Trend & Line-Item Tracking

TrendsView.tsx

- Sequential 6-month historical trends with visual sparkline indicators.
- Automatically calculates historical minimums, maximums, monthly averages, and variance trajectories per account line.

MODULE 7

Expense Composition Analysis

ExpenseCompositionView.tsx

- Evaluates the structural distribution of operating expenses.
- Computes each overhead category as a percentage of total revenue and total OPEX.
- Highlights shifting expense concentration to assist with cost rationalization.

- **Multi-Tier Model Cascade:** Employs gemini-3.1-flash-lite (primary) and gemini-3.8-flash (secondary) via the Google GenAI SDK.
- **Standard 6-Section Executive Report:** Executive Summary; Key Positive Movements; Key Negative Movements; Areas Requiring Attention (Red Flags); Possible Explanations; Recommended Management Actions.
- **Interactive Advisory Q&A:** Enables free-text queries (“Why did our gross margin fall in June?”, “Which 3 expenses grew the most?”) answered with certified citations.
- **Zero-Downtime Contingency Fallback:** If external cloud APIs are unavailable, an automated deterministic rules engine synthesizes a full commentary in under 50 milliseconds.

- **Executive Management Package:** Printable dashboard ready for board packs and executive committee reviews.
- **Advanced Excel Export (.xlsx):** Generates a 7-sheet workbook — Executive KPI Summary; Profit & Loss Statement; Variance & Materiality Analysis; Red Flags & Anomaly Register; Trend Analysis & Line-Item Statistics; Account Classification Mapping; Data Quality Audit Log.
- **Print & PDF Export:** One-click generation of formatted executive summary PDFs with Accuflow branding.

3. Technology Stack & Architecture

Layer	Technologies Used
Frontend UI	React 18, TypeScript, Tailwind CSS, Lucide React, Recharts
Server Backend	Node.js, Express.js (Port 3000), tsx (development), esbuild (production bundle)
AI / Intelligence	@google/genai TypeScript SDK, Gemini Flash Models, Fallback Deterministic Engine
Spreadsheet Processing	SheetJS (xlsx) for parsing and multi-sheet generation
Document Generation	jsPDF for client-side formatted PDF generation
Packaging & Launchers	Windows Batch Script (start_accuflow.bat), Windows Executable (Accuflow-CFO-Lens.exe)

4. Key Architectural & Security Decisions

- **Server-Side API Key Protection:** The GEMINI_API_KEY is strictly confined to the Express server environment. Browser clients communicate only with authenticated /api/* proxies.
- **Stateless Privacy Guarantee:** Client-uploaded financial spreadsheets are processed in-memory during the active session. No proprietary ledger records or customer data are written to external cloud databases.

- **Robustness Over Fragility:** All AI endpoints implement circuit-breaker fallbacks. If an external AI provider encounters latency spikes, users still receive immediate, mathematically accurate management reports.

This document is a formalized project summary of Accuflow Accounting Solutions CFO Lens, Release 1.0.0 Production, prepared for internal reference and stakeholder review.